

Report R22 — what the transition graph means

Weak components, terminal components and Hilbert-space fragmentation in open quantum systems

Tier 1 analysis

2026-08-18

Abstract

Every sector count this project reports is a count of components of one directed graph: the support graph of the channel in the computational basis. For unitary rules the choice of component notion is immaterial, and the project has been able to say “Krylov sector” without qualification. For dissipative rules it is not immaterial, and the two natural counts can disagree *exponentially*: we have rules with a single weakly connected component and ψ^N terminal components, $\psi = 1.4656$. This report says what the two counts are in the language of open quantum systems, and answers the question that provoked it — if the recurrent classes proliferate but the weak components do not, is that Hilbert-space fragmentation? The answer is **no**, and the reason is sharp rather than terminological. Weak components are exactly the blocks of a *strong symmetry* in the sense of Buča and Prosen: the finest decomposition of the basis that makes every Kraus operator block-diagonal, hence a genuine superselection rule with a charge measurable at $t = 0$. Terminal components are the *enclosures* of the Baumgartner–Narnhofer / Albert–Jiang asymptotic decomposition: they count steady states, not sectors. Made dimensional, the two counts are $\dim(\mathcal{C} \cap \mathcal{D}) = n_{\text{wcc}}$ and $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\text{rec}}$ for the strong-symmetry commutant $\mathcal{C}$ and the conserved-quantity space $\mathcal{F}$, so the graph inequality $n_{\text{rec}} \geq n_{\text{wcc}}$ is just the inclusion $\mathcal{C} \subseteq \mathcal{F}$, and Moudgalya–Motrunich’s commutant criterion applies verbatim to the first. The central result is that the gap between them is invisible to the *idempotents*: the projections in $\mathcal{F}$ are exactly the indicators of unions of weak components, so a channel may acquire ψ^N steady states without the Boolean algebra of always-answerable questions gaining a single atom. Fragmentation is exponential growth of that Boolean algebra; exponentially many attractors inside one block is multistability. Two hypotheses bound when the gap can occur at all: unital channels have no transient part (so all three component notions coincide), and functional support graphs have one attractor per component — **the gap requires the channel to be simultaneously non-unital and branching**, which 891 measured points across the 81 deterministic rules confirm without exception. Operationally, a strong symmetry says the attractor is a deterministic function of the initial state: measured, 100% of basis states have a certain fate for rules whose sectors proliferate, against 46% at $N = 10$ and 39% at $N = 12$ for a rule with one weak component. Yet the retained information $I(X;T)$ is extensive in *both* regimes, at comparable rates — so the difference is not how much is remembered but with what certainty: fragmentation is a noiseless code, multistability a noisy channel of similar capacity from which nothing at all can be recovered with probability one. We map the four growth regimes onto the literature, flag a terminology collision between two independent “strong/weak” dichotomies that this project uses both of, and state precisely what a population-level graph can and cannot certify.

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1 Why the question does not arise for closed systems

The object under study is the support graph G of one full brick-wall cycle Φ in the computational basis: vertices are the 2^N bit strings, and $x \rightarrow y$ when the channel can take $|x\rangle\langle x|$ to something with weight on $|y\rangle\langle y|$. Writing $\Phi(\rho) = \sum_j K_j \rho K_j^\dagger$, this is

$$x \rightarrow y \iff \exists j : \langle y | K_j | x \rangle \neq 0. \quad (1)$$

Three notions of “component” live on a directed graph: weakly connected components (WCCs, obtained by forgetting edge directions), strongly connected components (SCCs, mutual reachability), and terminal SCCs — those with no outgoing edge, which we also call recurrent classes or attractors. In general these are different. The following says why the distinction has never had to be made in the closed-system fragmentation literature.

Throughout, P denotes the induced population dynamics,

$$P_{yx} = \sum_j |\langle y | K_j | x \rangle|^2, \quad (2)$$

a column-stochastic matrix whose support is exactly G . Two structural hypotheses turn out to control everything, and neither is the one we first reached for.

Proposition 1 (Unitality kills the transient part). *If Φ is unital, $\Phi(\mathbb{I}) = \mathbb{I}$, then P is doubly stochastic, no state is transient, and*

$$WCC = SCC = \text{terminal SCC}.$$

Unitary evolution is the special case $\Phi(\rho) = U\rho U^\dagger$.

Proof. Columns of P sum to one by trace preservation, $\sum_j K_j^\dagger K_j = \mathbb{I}$. For the rows, $\sum_x P_{yx} = \sum_j \langle y | K_j K_j^\dagger | y \rangle = \langle y | \Phi(\mathbb{I}) | y \rangle$, which is 1 exactly when Φ is unital on the diagonal. So P is doubly stochastic, the uniform distribution is stationary with full support, and since a transient state of a finite chain carries zero stationary mass, there are none; every SCC is therefore terminal. Finally, distinct terminal SCCs admit no edge in either direction — an edge leaving a terminal SCC contradicts terminality — so each is its own WCC. $\square$

Stating it for unital rather than unitary channels is not generality for its own sake: it identifies the physical ingredient. Unitality says the maximally mixed state is stationary, i.e. the channel has no contracting direction in state space. A Hadamard branches but does not contract; a reset contracts. That is why the phenomenon of this report is invisible until a D or an E enters the rule word. Empirically the dichotomy is exact: across all 16 unitary rules and all $N \in [6, 16]$ the measured transient fraction is identically 0 and $n_{\text{rec}} = n_{\text{wcc}}$ at every point, while no dissipative rule has an empty transient part at any N ; measured row sums of P deviate from 1 by as much as 15 for W_{36} at $N = 8$.

Contraction alone is not enough, however. The second hypothesis is branching.

Proposition 2 (Determinism also collapses the two counts). *If every vertex of G has out-degree one — the support graph is a functional graph — then every WCC contains exactly one terminal SCC, so $n_{\text{rec}} = n_{\text{wcc}}$ regardless of how contracting Φ is.*

Proof. A weakly connected component of a functional graph on n vertices has exactly n edges, hence exactly one cycle; following the unique successor from any vertex must reach it. That cycle is the unique terminal SCC of the component. $\square$

Together the two propositions bound the phenomenon from both sides:

A gap between n_{wcc} and n_{rec} requires the channel to be simultaneously non-unital and branching.

Both halves are visible in our rule family. The 81 rules whose words contain no V have out-degree one, and across all 891 measured (rule, N) points among them $n_{\text{rec}} = n_{\text{wcc}}$ without a single exception. The 16 rules with no D or E are unital, and satisfy the same equality by Proposition 1. Every rule for which the two counts actually separate therefore carries at least one V and at least one reset — and, as it happens, all eight carry exactly one V , the minimum amount of branching that permits the gap at all. **The subject of this report is what happens when branching and contraction act together, and it has no closed-system shadow.**

2 The dictionary

2.1 Weak components are a strong symmetry

Proposition 3 (WCCs block-diagonalise every Kraus operator). *Let $\{B_\alpha\}$ be the WCCs of G and P_α the orthogonal projector onto $\text{span}\{|x\rangle : x \in B_\alpha\}$. Then $[K_j, P_\alpha] = 0$ for every Kraus operator K_j and every α , and consequently $\text{Tr}(P_\alpha \rho)$ is conserved for all time. Conversely, the $\{B_\alpha\}$ are the finest partition of the computational basis with this property.*

Proof. If $P_\alpha K_j P_\beta \neq 0$ for $\alpha \neq \beta$ there are $x \in B_\beta$, $y \in B_\alpha$ with $\langle y | K_j | x \rangle \neq 0$, i.e. an edge $x \rightarrow y$ in (1); then x and y are weakly connected and $\alpha = \beta$, a contradiction. Hence each $K_j = \bigoplus_\alpha P_\alpha K_j P_\alpha$ and $P_\alpha K_j = K_j P_\alpha$. Conservation follows: $\text{Tr}(P_\alpha \Phi(\rho)) = \sum_j \text{Tr}(K_j P_\alpha \rho K_j^\dagger) = \sum_j \text{Tr}(K_j^\dagger K_j P_\alpha \rho) = \text{Tr}(P_\alpha \rho)$, using $\sum_j K_j^\dagger K_j = \mathbb{I}$. Minimality holds because any coarser-than- G partition with block-diagonal Kraus operators has no edges between its parts, so it is a union of WCCs. $\square$

Proposition 3 is the whole content of the identification. In the classification of Buča and Prosen [1], a symmetry of a Lindbladian is *strong* when the Hamiltonian and every jump operator individually commute with it, and *weak* when only the generator as a whole does. Setting $U = \sum_\alpha e^{i\theta_\alpha} P_\alpha$ for generic phases, Proposition 3 says exactly that the WCC decomposition furnishes a strong symmetry, and the minimality clause says it is the maximal one available in

the computational basis. Buča and Prosen establish the fact that gives this report its answer: *only* a strong symmetry implies non-trivial reductions inside the space of steady states.

The statement can be made dimensional, which is what connects it to the commutant criterion. Write $\mathcal{C} = \{A : [A, K_j] = 0 \ \forall j\}$ for the commutant of the Kraus set — the algebra of strong symmetries — and $\mathcal{D}$ for the algebra of operators diagonal in the computational basis.

Proposition 4 (n_{wcc} is a commutant dimension). $\dim(\mathcal{C} \cap \mathcal{D}) = n_{\text{wcc}}$. *Concretely, $\mathcal{C} \cap \mathcal{D}$ consists exactly of the operators $A = \sum_x a(x)|x\rangle\langle x|$ whose symbol a is constant on each weak component.*

Proof. For diagonal A , $[A, K_j] = 0$ reads $(a(y) - a(x)) \langle y|K_j|x\rangle = 0$ for all x, y, j , i.e. $a(y) = a(x)$ whenever $x \rightarrow y$. Since a is a scalar function, the condition is symmetric under reversing the edge, so a is constant on weakly connected components — and any such a manifestly commutes with every K_j . $\square$

This is the precise bridge to Moudgalya and Motrunich [5], whose criterion for fragmentation is exponential growth of the dimension of the commutant algebra, with conventional symmetries such as $U(1)$ or $SU(2)$ giving only polynomial growth. Proposition 4 says that n_{wcc} is not merely *correlated* with that dimension; in the classical (product-basis) setting it *is* that dimension. So:

n_{wcc} counts superselection sectors, and equals the dimension of the diagonal strong-symmetry commutant. It is the open-system successor of “number of Krylov sectors”, and it is the count that fragmentation is about.

2.2 Terminal components are enclosures

The terminal SCCs are a different kind of object entirely. They are the *enclosures* of Baumgartner and Narnhofer [2], whose analysis of GKS–Lindblad generators equips the Hilbert space with a decomposition into mutually orthogonal subspaces and separates the transient part from the part where probability accumulates. In the quantum-channel formulation the same structure is the irreducible decomposition of Carbone and Pautrat [4]: the recurrent subspace, which supports the invariant states, decomposes into the supports of the extremal invariant states, with the orthogonal complement being transient. Albert and Jiang [3] give the form of the asymptotic manifold,

$$\rho(\infty) = \bigoplus_k (\sigma_k \otimes \rho_k), \quad \sigma_k \in M_{n_k}, \quad (3)$$

where k runs over the enclosures, ρ_k is a fixed state on the “dissipative” factor of enclosure k , and M_{n_k} is a noiseless subsystem carrying surviving coherence. Their central point — and the one we lean on in §3 — is that the infinite-time state is fixed by the initial state together with the full list of conserved quantities.

So:

n_{rec} counts steady states. It is a statement about the $t \rightarrow \infty$ limit, not about a decomposition of the Hilbert space.

2.3 The table

graph object	algebraic object	physical reading
WCC	block of a strong symmetry; $[K_j, P_\alpha] = 0$	superselection sector; Krylov sector of the dephased model
n_{wcc}	# blocks	# conserved charges valid at <i>all</i> times
$d_{\text{max}}^{\text{wcc}}$	largest block dimension	the R21 strong/weak-fragmentation ratio
SCC	communicating class	refines the sector into transient strata
transient SCCs	transient subspace	everything that decays
terminal SCC	enclosure [2]; irreducible component [4]	attractor; support of one extremal steady state
n_{rec}	# enclosures	# extremal steady states
$d_{\text{max}}^{\text{rec}}$	$n_k \cdot m_k$ of the largest enclosure	noiseless-subsystem dimension <i>times</i> mixing dimension

Two structural facts follow immediately and are worth stating because they constrain every census. First, $n_{\text{rec}} \geq n_{\text{wcc}}$ always, since every weak component contains at least one terminal SCC; the inequality is saturated exactly when each sector has a unique attractor. Second, the familiar hyperbola $ab \geq 2$ of R9/R16 is a statement about WCCs *only*, and it holds because they tile: $\sum_\alpha \dim B_\alpha = 2^N$. Recurrent classes do not tile — they occupy the recurrent volume $\mu^N \leq 2^N$ — so the correct analogue is $a_{\text{rec}} b_{\text{rec}} \geq \mu$, with $\mu = 2$ recovering the closed-system bound. For the rules discussed below $\mu = \psi = 1.4656$ and the bound is saturated exactly.

3 Is exponential n_{rec} without exponential n_{wcc} fragmentation?

3.1 The criterion

The sharpest available definition of fragmentation is the commutant-algebra one of Moudgalya and Motrunich [5]: given the set of local terms (Hamiltonian terms, or the local gates of a circuit), form the algebra of all operators commuting with each of them; the system is fragmented when the dimension of this *commutant* grows exponentially in system size, as against the polynomial growth produced by conventional symmetries such as $U(1)$ or $SU(2)$. Their framework also supplies the classical/quantum distinction — “classical” fragmentation being fragmentation in a product basis, which is the case here.

The open-system version replaces “commutes with each local term” by “commutes with the Hamiltonian and each jump operator”, which is Buča–Prosen strong symmetry, and the superoperator formulation is developed in [6], where symmetry algebras appear as frustration-free ground states of a local super-Hamiltonian and the strong/weak distinction is made explicit. Under that criterion the answer is immediate: Proposition 3 identifies the strong-symmetry commutant in the computational basis with the WCC decomposition, so **exponentially many weak components is fragmentation and exponentially many terminal components is not**. A rule with one weak component has a trivial strong-symmetry commutant: its Hilbert space is a single Krylov sector.

3.2 What is exponentially large instead

This does not make the phenomenon empty — something is exponentially large, and it is worth naming precisely. It is the space of conserved quantities in the Heisenberg picture. For each terminal component T_i define the *absorption probability*

$$\varphi_i(x) = \Pr[\text{absorbed in } T_i \mid \text{start in } x], \quad J_i = \sum_x \varphi_i(x) |x\rangle\langle x|. \quad (4)$$

Each J_i satisfies $\Phi^\dagger(J_i) = J_i$, so $\text{Tr}(J_i \rho(t))$ is constant: these are conserved quantities in exactly Albert and Jiang's sense. Writing $\mathcal{F} = \{A : \Phi^\dagger(A) = A\}$ for the space of them, the counterpart of Proposition 4 holds.

Proposition 5 (n_{rec} is also a dimension, of a larger space). $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\text{rec}}$, with basis $\{J_i\}$. Moreover $\mathcal{C} \subseteq \mathcal{F}$, so $n_{\text{wcc}} \leq n_{\text{rec}}$ always.

Proof. A diagonal A with symbol a is fixed by $\Phi^\dagger$ iff a is harmonic, $a(x) = \sum_y P_{yx} a(y)$. By the maximum principle a is constant on each terminal component, and its values there determine it everywhere by solving the Dirichlet problem on the transient part; the solutions are spanned by the φ_i . The inclusion is immediate: $[A, K_j] = 0$ gives $\Phi^\dagger(A) = \sum_j K_j^\dagger A K_j = A \sum_j K_j^\dagger K_j = A$. $\square$

So the graph inequality $n_{\text{rec}} \geq n_{\text{wcc}}$ is nothing but the algebra inclusion $\mathcal{C} \subseteq \mathcal{F}$: *every strong symmetry is a conserved quantity, but not every conserved quantity is a strong symmetry*. Exponentially many attractors therefore do imply exponentially many conserved quantities. The question is what *kind* — and the answer is that the two spaces differ in a way no amount of multistability can repair.

Theorem 1 (The idempotents never proliferate). *The projections lying in $\mathcal{F} \cap \mathcal{D}$ are exactly the indicators of unions of weak components. Hence $\mathcal{F} \cap \mathcal{D}$ contains precisely $2^{n_{\text{wcc}}}$ projections, however large n_{rec} may be, and its minimal projections are the P_α of Proposition 3. In particular $J_i^2 = J_i$ if and only if T_i is the unique terminal component of its weak component, and $J_i^2 = J_i$ for every i if and only if $n_{\text{rec}} = n_{\text{wcc}}$.*

Proof. Let $Q = \sum_{x \in S} |x\rangle\langle x|$ be harmonic. For $x \in S$, $1 = \sum_{y \in S} P_{yx}$ forces every successor of x into S ; for $x \notin S$, $0 = \sum_{y \in S} P_{yx}$ forbids any successor of x from lying in S . Thus S and its complement are both forward-closed, so no edge joins them in either direction and S is a union of weak components. Conversely every such indicator is harmonic. The count and the statement about minimal projections follow, and the J_i clause is the special case $S = \text{supp } \varphi_i$. $\square$

Theorem 1 is the whole answer, and it is stronger than the observation that non-idempotent conserved quantities exist. *The idempotent part of the conserved-quantity space is blind to the attractors entirely.* A rule may acquire ψ^N steady states, and the Boolean algebra of yes/no questions about the initial state that remain answerable at all times does not grow by a single atom. Fragmentation is exponential growth of that Boolean algebra; multistability is exponential growth of the surrounding linear space and nothing more. Measured directly at $N = 8$: W_{73} has 19 idempotent J_i out of 19, W_{36} has 9 out of 28 — one for each of the nine weak components that happen to hold a single attractor — and W_{203} , whose entire state space is one weak component, has *none*.

3.3 The operational content, measured

Proposition 1 converts the abstract distinction into a number one can compute: what fraction of basis states have their attractor determined *with certainty* by the initial condition? We built the honest population Markov chain of the channel (I keeps, $V = H$ splits $\frac{1}{2}/\frac{1}{2}$, D and E reset) and solved $\varphi = Q\varphi + R$ exactly on the transient block.

rule	regime	n_{wcc}	n_{rec}	certain fate	effective # attractors
W_{73} VIID	sectors proliferate	41	41	100.0%	1.00
W_{109} EIIIV	sectors proliferate	54	54	100.0%	1.00
W_{36} IDDV	attractors only	10	60	72.5%	1.8
W_{203} VEII	attractors only	1	28	46.1%	2.6
W_{203} VEII ($N = 12$)	attractors only	1	60	39.1%	3.2

All rows at $N = 10$ unless marked, obc0. “Effective # attractors” is the participation ratio $1/\sum_i \varphi_i(x)^2$ averaged over basis states. The dissipative rules whose *weak* components proliferate behave exactly as Proposition 1 requires: every basis state has a certain fate, the conserved quantities are projectors, the charge is fixed at $t = 0$. W_{203} , with a single weak component, does not: fewer than half its basis states have a determined fate, and the fraction *falls* with N while the effective number of accessible attractors rises (max 15.0 at $N = 10$, 27.1 at $N = 12$).

The physical statement is therefore:

With exponentially many weak components, **the initial state decides** which attractor the system reaches. With one weak component and exponentially many attractors, **the noise decides**; the initial state only sets the odds.

Both are ergodicity breaking, and both give an exponentially degenerate steady-state manifold. Only the first is Hilbert-space fragmentation.

3.4 How much memory survives, in bits

Theorem 1 says the *certain* memory does not grow. It does not say the memory is small, and it would be wrong to conclude that. Let X be a uniformly random initial basis state and T the terminal component the system ends in. The natural quantities are the entropy of the outcome $H(T)$, the noise $H(T | X)$ injected by the branching, and the retained information $I(X; T) = H(T) - H(T | X)$.

rule	regime	N	$H(T)$	$H(T X)$	$I(X; T)$	ΔI per 2 sites
W_{73} VIID	sectors proliferate	8	3.170	0.000	3.170	—
		12	4.832	0.000	4.832	0.831
W_{109} EIIIV	sectors proliferate	12	5.471	0.000	5.471	0.831
W_{36} IDDV	attractors only	12	6.588	0.710	5.878	0.889
W_{203} VEII	attractors only	8	3.626	0.805	2.821	—
		12	5.786	1.242	4.544	0.862
W_{219} VEEI	attractors only	12	4.603	1.070	3.533	0.886

All in bits, obc0, X uniform. Three things are worth reading off.

First, $H(T | X) = 0$ identically in the top block, which is Theorem 1 again in information-theoretic dress: when every φ_i is idempotent the outcome is a deterministic function of the input and the channel from X to T is noiseless.

Second, $H(T | X) > 0$ and *growing* in the bottom block — for W_{203} it rises $0.805 \rightarrow 1.024 \rightarrow 1.242$ across $N = 8, 10, 12$, an extensive loss of roughly 0.11 bits per site. A constant fraction of the capacity of this channel is consumed by noise, and that fraction does not diminish with system size.

Third, and least expected: $I(X; T)$ is *extensive in both regimes*, with strikingly similar coefficients — the increment per two sites is 0.831 for the fragmented rules and 0.862–0.889 for the multistable ones. Case B is therefore not a weak or vestigial form of memory. It stores an extensive number of bits about the initial condition, comparable in count to what genuine fragmentation stores.

The distinction survives all of this, and sharpens: it is not *how much* is remembered but *with what certainty*. In the fragmented case the map $x \mapsto T$ is a deterministic labelling, so there is a decoder that is correct with probability one and is itself a conserved quantity. In the multistable case no such decoder exists — by Theorem 1 the only zero-error questions are the WCC memberships, and there is just one weak component, so **nothing whatever about the initial state can be recovered with certainty**, even though $\sim 0.38N$ bits can be recovered on average. Fragmentation gives a noiseless code; multistability gives a noisy channel of comparable rate.

3.5 Where this sits in the current literature

Two neighbouring bodies of work make the placement precise.

Dissipative freezing. Sánchez Muñoz, Buca and co-workers [10, 11] study what individual quantum trajectories do under a *strong* symmetry: initialised in a superposition of sectors, each stochastic realisation randomly selects one and stays there, so the conservation law that holds for the density matrix is “broken” at the level of single realisations. Our case is the mirror image. There, trajectory selection happens *despite* a strong symmetry that makes the ensemble remember; here there is no strong symmetry at all, and the trajectory-level selection among ψ^N attractors is the only structure present. The two phenomena are easy to confuse from the trajectory picture alone and are distinguished exactly by whether φ_i is idempotent.

Dark states. When the attractors are single basis states — the generic case among the rules where this gap appears — the terminal components are dark states in the standard sense: configurations immune to every jump operator, which is precisely how the recurrent subspace is characterised in open reaction–diffusion settings. An exponentially large dark-state manifold reached from a structureless transient bulk is a well-recognised target for dissipative state preparation, and it is a different resource from a fragmented Hilbert space: one prepares states, the other protects them.

4 The four regimes

Writing $n_{\text{wcc}} \sim a_{\text{w}}^N$, $n_{\text{rec}} \sim a_{\text{r}}^N$, $d_{\text{max}}^{\text{rec}} \sim b_{\text{r}}^N$, four regimes are possible and each has a distinct reading.

(A/D) Sectors, attractors and attractor dimension all exponential. Exponentially many superselection sectors, each retaining an exponentially large recurrent space. This is fragmentation in the full sense, and it is the regime in which the asymptotic manifold (3) can carry an exponentially large noiseless subsystem — strong fragmentation is a known route to stabilising exponentially many decoherence-free subspaces. Whether $d_{\text{max}}^{\text{rec}} = n_k m_k$ is coherence (n_k) or classical mixing (m_k) is *not* decided by the graph; see §6. In our setting the unitary rules W_{108} , W_{201} , W_{156} , W_{198} sit here together with the dissipative W_{73} , W_{109} .

(A/C) Sectors exponential, attractors trivial. Exponentially many sectors, each collapsing onto a small attractor — typically one frozen state per sector. The fragmentation is real and the memory is classical: the long-time state is a bit string labelled by the conserved charge. This is what dephasing does to a fragmented model, and it is the regime Li, Sala and Pollmann [7] reach when they couple a fragmented system to a dephasing bath: the quantum fragmentation is reduced to a classical one, the stationary state becomes separable, and what survives is classical correlations arising from fluctuations of the remaining conserved quantities. The degenerate limiting case is $W_{204} = \text{IIII}$, with 2^N sectors of one state each.

(B/C) Attractors exponential, sectors not, attractors trivial. The regime that provoked this report. One (or polynomially many) Krylov sectors, ψ^N dark states, a transient bulk occupying 99%+ of the Hilbert space, and non-idempotent conserved quantities. Not fragmentation: **multistability with an exponentially degenerate dark-state manifold.** The exemplar is $W_{203} = \text{VEII}$, whose entire 2^N -dimensional space is one weak component.

(B/D) Attractors exponential and exponentially large, sectors not. Exponentially many enclosures each exponentially large, inside a single superselection sector. This would be the most interesting regime — an exponentially degenerate manifold of exponentially large noiseless subsystems with no symmetry organising them — and it is *empty* in the 256-rule family

at obc0. We record its absence as an observation, not a theorem; nothing we know forbids it, and the volume constraint $a_r b_r \leq 2$ leaves room for it.

5 A terminology collision worth flagging

This project now uses two independent dichotomies that are both called “strong versus weak”, and they are orthogonal. Conflating them would invalidate R21’s conclusions or this report’s, depending on the direction of the error.

strong / weak <i>fragmentation</i> Sala <i>et al.</i> [8], Khemani <i>et al.</i> [9]	strong / weak <i>symmetry</i> Buča–Prosen [1]
About the <i>size</i> of the largest sector relative to the symmetry sector containing it: strong iff $D_{\max}(s)/\dim s \rightarrow 0$.	About <i>how</i> a symmetry acts: strong iff every jump operator commutes with it individually, weak iff only the generator does.
A question you ask <i>after</i> you have a sector decomposition.	A question about whether you have a sector decomposition at all.
Used in R21, which grades 108/201/156/198 as strongly fragmented.	Used here, to say what n_{wcc} counts.

A model can be weakly fragmented with a perfectly good strong symmetry (W_{60} , W_{102} in R21), and — as §3 shows — can have exponentially many steady states with no strong symmetry whatever. There is also a third, newer use of the same words in the mixed-state literature: *strong-to-weak spontaneous symmetry breaking* [12], where a strong symmetry of a density matrix degrades to a weak one and the order parameter is a fidelity correlator rather than a two-point function. That is a statement about phases of mixed states, not about sector counting, and it should not be read into either column above. Whether the boundary between our regimes A and B can be crossed as a genuine SWSSB transition in a family of channels is a natural question and one we have not touched.

6 What a population graph cannot certify

The support graph (1) is the *population* skeleton of the channel. It records which basis states feed which, and nothing about the phases. Three consequences bound every claim in this report and in the project’s dissipative reports generally.

1. **It is a lower bound on fragmentation.** A channel may have conserved quantities invisible in the computational basis — the fragmentation may be quantum, i.e. in a non-product basis, in the sense of [5]. The WCC count is then an undercount. This is not hypothetical for us: R-T15 identified the single- V commutant of $W_{108}/W_{201}/W_{156}/W_{198}$ as a Temperley–Lieb algebra, and the Temperley–Lieb model is exactly the vehicle Li, Sala and Pollmann use for *quantum* HSF in an entangled basis [7]. Their fragmentation-preserving coupling produces a steady state with coherent memory carried by an extensive set of *non-commuting* conserved quantities. No diagonal graph can see any of that.
2. **It cannot split $d_{\max}^{\text{rec}}$ into $n_k \cdot m_k$.** The graph gives the dimension of an enclosure’s support; (3) says that dimension factorises into a noiseless subsystem and a mixing factor, and which one carries the growth is a question about coherence. The project has answered it in both directions already: R17 found the exponential terminal SCCs to be dense, mixing digraphs rather than cycles — so m_k is large and n_k small, a classical attractor — while the X -gate correction established the opposite for permutation circuits, where the functional graph describes only populations and exponentially large decoherence-free subspaces sit underneath it. *The graph alone never settles this.*

3. **The equality $\dim(\text{fixed points of } \Phi^\dagger) = n_{\text{rec}}$ is a classical statement.** For the population chain the conserved quantities are exhausted by the absorption probabilities (4). The full channel can carry additional fixed points built from coherences between enclosures of equal structure, which is precisely the M_{n_k} factor in (3). So n_{rec} is a lower bound on the steady-state count too.

7 Summary

- WCCs are the blocks of a *strong symmetry*: the finest basis decomposition making every Kraus operator block-diagonal (Prop. 3), and $n_{\text{wcc}} = \dim(\mathcal{C} \cap \mathcal{D})$ exactly (Prop. 4), which is the quantity Moudgalya–Motrunich’s criterion is stated in terms of. Exponential growth of it is Hilbert-space fragmentation.
- Terminal SCCs are *enclosures*: supports of extremal steady states, with $n_{\text{rec}} = \dim(\mathcal{F} \cap \mathcal{D})$ (Prop. 5). The familiar inequality $n_{\text{rec}} \geq n_{\text{wcc}}$ is the algebra inclusion $\mathcal{C} \subseteq \mathcal{F}$.
- Exponentially many terminal SCCs inside one WCC is **not** fragmentation. The projections in $\mathcal{F}$ are exactly the unions of weak components (Thm. 1), so the idempotent part of the conserved-quantity space cannot see the extra attractors at all; the attractor is then selected by noise rather than fixed by the initial state — a 100% versus 46% effect in direct measurement (§3.3).
- But the memory is not weak, only uncertain: $I(X; T)$ is extensive in both regimes at comparable rates (§3.4). Fragmentation is a noiseless code; multistability is a noisy channel of similar capacity whose zero-error capacity is nil.
- The gap requires the channel to be **both non-unital and branching**: unital channels have no transient part (Prop. 1) and functional graphs have one attractor per component (Prop. 2). Hence no closed-system analogue, and in this family all eight rules exhibiting the gap carry exactly one V together with at least one reset.
- “Strong/weak” means two different things in this project, and now a third thing in the mixed-state literature (§5).
- Everything here is certified at the level of populations only (§6); coherent fragmentation would be invisible, and our own Temperley–Lieb structure is a live candidate for it.

Reproduce

The measurement of §3.3 needs transition *probabilities*, which the project’s component engine does not carry: `graph/scc.py` works with the unweighted successor relation, since that is all that component structure requires. The weighted one-cycle chain is therefore rebuilt from the gate semantics (I keeps, $V = H$ splits $\frac{1}{2}/\frac{1}{2}$, D and E reset), in `scaling/absorption.py`, at cost $O(4^N)$. It is a small- N diagnostic, not a sweep.

```
python -m qca_fragmentation.scaling.absorption --table
pytest qca_fragmentation/tests/test_absorption.py    # 101 tests
```

The tests pin every number quoted in §3.3 and §3.4; check both directions of Theorem 1, including the idempotent counts 19/19, 9/28 and 0/13 and the bound $\#\{\text{idempotents}\} \leq n_{\text{wcc}}$; verify the double stochasticity of Proposition 1 (holding for the unitary rules, failing for the dissipative ones); and confirm Proposition 2 on the V -free rules. The extensivity of $I(X; T)$ is checked as an equality of successive increments to 5%.

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